

PROJECT REPORT OF INDUSTRY ORIENTED HANDS-ON EXPERIENCE (IOHE)

ON

**An Efficient CNN-Based Framework for Automated Blood Cell Classification in
Hematological Disease Diagnosis**

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**An Efficient CNN-Based Framework for Automated Blood Cell Classification in Hematological Disease Diagnosis**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Rajat Takkar. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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The research work and implementation were carried out using the Jupyter Notebook platform with the support of Python libraries such as NumPy, Pandas, Matplotlib, Seaborn, and Scikit-learn, along with deep learning frameworks used for developing the CNN-based blood cell classification model.

We would also like to acknowledge the use of the ALL Challenge dataset of ISBI 2019 (C-NMC 2019) obtained from Kaggle and The Cancer Imaging Archive, which served as the foundation for training and evaluating the proposed model for automated blood cell classification.

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1. ABSTRACT

This research focuses on the automated classification of blood cell images using a Convolutional Neural Network (CNN)-based deep learning framework for hematological disease diagnosis. The proposed system classifies microscopic blood cell images into four categories: Benign, Malignant Pre-B, Malignant Pro-B, and Malignant Early Pre-B.

The dataset used in this study is the ALL Challenge dataset of ISBI 2019 (C-NMC 2019), containing 3,242 microscopic blood cell images collected from Kaggle. Image preprocessing techniques such as resizing, normalization, and enhancement are applied to improve the quality of the input data before training the CNN model.

The proposed CNN architecture automatically extracts important morphological features from blood cell images and performs accurate classification of benign and malignant cells. The model achieved an overall classification accuracy of 98.76% along with high precision, recall, and F1-score values across all classes. Performance evaluation was further validated using training and validation accuracy curves, loss analysis, and confusion matrix evaluation.

The proposed framework provides a fast, reliable, and automated alternative to manual microscopic blood cell analysis, reducing diagnostic time and minimizing human error. This system can support medical professionals in the early detection and diagnosis of hematological diseases such as leukemia and improve the efficiency of clinical decision-making.

2. INTRODUCTION

Blood cell classification plays an important role in the diagnosis and treatment of hematological diseases such as leukemia, anemia, and lymphoma. Blood mainly consists of red blood cells (RBCs), white blood cells (WBCs), and platelets, where white blood cells are especially significant for detecting abnormalities related to blood cancer. Traditionally, the examination and classification of blood smear images are performed manually by medical experts using microscopes. However, manual analysis is time-consuming, subjective, and highly dependent on the experience of pathologists, which may lead to diagnostic errors and inconsistencies. Therefore, there is a growing need for automated and intelligent systems that can accurately classify blood cells and assist medical professionals in disease diagnosis.

This research focuses on automated blood cell classification using a Convolutional Neural Network (CNN)-based deep learning framework. The primary objective of this work is to classify microscopic blood cell images into four clinically important categories: Benign, Malignant Pre-B, Malignant Pro-B, and Malignant Early Pre-B. Accurate identification of these categories is essential for the early detection and diagnosis of leukemia and other hematological disorders.

Deep learning techniques, especially CNNs, have shown remarkable success in medical image analysis because of their ability to automatically learn complex visual and morphological features from images. Unlike traditional image-processing methods that require manual feature extraction, CNNs can directly learn meaningful patterns such as cell shape, texture, boundaries, and internal structures from blood smear images. This makes CNNs highly effective for automated blood cell classification tasks.

In this study, the ALL Challenge dataset of ISBI 2019 (C-NMC 2019) obtained from Kaggle is used for training and testing the proposed model. The dataset contains 3,242 microscopic blood cell images divided into four classes. Various preprocessing techniques such as image resizing, normalization, and enhancement are applied to improve image quality and ensure consistent input for the CNN model.

The implementation of the proposed framework includes data preprocessing, CNN model development, feature extraction, model training, and performance evaluation. Different evaluation metrics such as accuracy, precision, recall, F1-score, confusion matrix, and training-validation curves are used to measure the effectiveness of the model. The proposed CNN model achieved high classification accuracy and demonstrated strong capability in distinguishing benign and malignant blood cell patterns.

Overall, this research demonstrates how deep learning and CNN-based techniques can be effectively applied in the field of medical image analysis to improve blood cell classification and hematological disease diagnosis. The proposed automated framework can assist healthcare professionals by reducing diagnostic time, minimizing human error, and providing reliable clinical decision support for early disease detection.

3. METHODOLOGY

The methodology of the proposed research consists of several important stages for automated blood cell classification using a Convolutional Neural Network (CNN)-based framework. The overall process includes dataset collection, image preprocessing, CNN model development, training, and performance evaluation.

3.1 Dataset Collection

The dataset used in this research is the ALL Challenge dataset of ISBI 2019 (C-NMC 2019) obtained from Kaggle and The Cancer Imaging Archive.

The dataset contains 3,242 microscopic blood cell images categorized into four classes:

- Benign
- Malignant Pre-B
- Malignant Pro-B
- Malignant Early Pre-B

The dataset is used for training, validation, and testing of the proposed CNN model.

3.2 Image Preprocessing

Before training the CNN model, the blood cell images are preprocessed to improve image quality and ensure consistency. The preprocessing steps include:

- Image resizing to a fixed dimension
- Image normalization for better pixel value distribution
- Image enhancement to improve visibility of cell structures
- Data organization into different class folders

These preprocessing techniques help the CNN model learn meaningful morphological features effectively.

3.3 Data Visualization

Visualization techniques are used to analyze the dataset and understand the distribution of blood cell classes. Graphs and plots are generated using Matplotlib and Seaborn to study:

- Distribution of blood cell classes
- Sample microscopic blood cell images
- Training and validation accuracy curves
- Training and validation loss curves
- Confusion matrix analysis

These visualizations help evaluate the learning behavior and performance of the proposed CNN model.

3.4 CNN Model Development

A Convolutional Neural Network (CNN) model is developed for automated blood cell classification. The architecture mainly consists of:

- Convolutional layers for feature extraction
- Max-pooling layers for dimensionality reduction
- Batch normalization layers for stable training
- Dropout layers to reduce overfitting
- Fully connected dense layers for classification
- SoftMax activation function for final class prediction

The CNN automatically learns important image features such as cell shape, texture, boundaries, and internal structures from microscopic blood cell images.

3.5 Model Training and Classification

The processed dataset is used to train the CNN model. During training, the model learns to classify blood cell images into the following four categories:

- Benign
- Malignant Pre-B
- Malignant Pro-B
- Malignant Early Pre-B

The model is trained over multiple epochs using optimized parameters to improve classification accuracy and generalization performance.

3.6 Model Evaluation

The performance of the proposed CNN model is evaluated using different classification metrics, including:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Training and validation accuracy/loss curves are also analyzed to measure model stability and detect overfitting. The proposed model achieved an overall classification accuracy of 98.76%, demonstrating its effectiveness in automated blood cell classification and hematological disease diagnosis.

4. TOOLS AND TECHNOLOGIES

This research uses a combination of deep learning, image processing, and data visualization tools to develop an efficient CNN-based framework for automated blood cell classification and hematological disease diagnosis.

Programming Language

- Python is used as the primary programming language because of its simplicity, flexibility, and strong support for deep learning, medical image analysis, and machine learning applications.

Libraries and Frameworks

- NumPy → used for numerical computations and array operations during image processing and model implementation
- Pandas → used for data handling, dataset organization, and preprocessing tasks
- Matplotlib → used for plotting graphs such as accuracy curves, loss curves, and confusion matrix visualization
- Seaborn → used for advanced data visualization and performance analysis
- Scikit-learn → used for performance evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrix generation
- TensorFlow and Keras → used for building, training, and evaluating the Convolutional Neural Network (CNN) model

Deep Learning Techniques

- Convolutional Neural Network (CNN) → used for automated feature extraction and blood cell image classification
- Convolutional Layers → used for learning important image features such as texture, shape, and cell boundaries
- Max-Pooling Layers → used for reducing spatial dimensions and computational complexity
- Dropout Layers → used to reduce overfitting and improve model generalization
- SoftMax Activation Function → used for multi-class blood cell classification

Dataset and Image Processing

- ALL Challenge dataset of ISBI 2019 (C-NMC 2019) → used for training and testing the proposed blood cell classification framework
- Image preprocessing techniques such as resizing, normalization, and enhancement are applied to improve image quality and model performance

Development Environment

- Jupyter Notebook → used as the development environment for coding, training the CNN model, visualization, and performance analysis

5. IMPLEMENTATION

The implementation of the proposed research is divided into several stages for developing an automated blood cell classification system using a Convolutional Neural Network (CNN). The complete implementation process includes dataset preparation, image preprocessing, CNN model development, training, visualization, and performance evaluation.

5.1 Dataset Preparation

The dataset used in this research is the ALL Challenge dataset of ISBI 2019 (C-NMC 2019) obtained from Kaggle.

The dataset contains 3,242 microscopic blood cell images divided into four classes:

- Benign
- Malignant Pre-B
- Malignant Pro-B
- Malignant Early Pre-B

The images are organized into separate folders according to their class labels for supervised learning.

5.2 Image Preprocessing

Before training the CNN model, several preprocessing techniques are applied to improve the quality and consistency of the input images. These steps include:

- Image resizing to fixed dimensions
- Image normalization for balanced pixel intensity distribution
- Image enhancement for better visibility of cell structures
- Dataset splitting into training, validation, and testing sets

These preprocessing steps help the CNN model learn discriminative features more effectively.

5.3 CNN Architecture Implementation

A Convolutional Neural Network (CNN) architecture is implemented for blood cell classification.

The architecture consists of:

- Convolutional layers for feature extraction
- Max-pooling layers for reducing feature dimensions
- Batch normalization layers for stable learning
- Dropout layers for preventing overfitting
- Fully connected dense layers for classification
- SoftMax activation function for multi-class prediction

The CNN automatically extracts important morphological features such as cell texture, shape, boundaries, and internal structures from microscopic blood cell images.

5.4 Data Visualization

Different visualization techniques are implemented using Matplotlib and Seaborn for performance analysis and interpretation. The visualizations include:

- Sample blood cell image visualization
- Blood cell class distribution graphs
- Training and validation accuracy curves
- Training and validation loss curves
- Confusion matrix visualization

These plots help analyze model learning behavior and classification performance.

5.5 Model Training

The CNN model is trained using the processed blood cell image dataset over multiple epochs.

During training:

- The model learns discriminative image features automatically
- Training accuracy and loss are continuously monitored
- Validation performance is analyzed to avoid overfitting
- Optimized parameters are used to improve classification performance

The trained CNN model classifies blood cell images into the following categories:

- Benign
- Malignant Pre-B
- Malignant Pro-B
- Malignant Early Pre-B

5.6 Performance Evaluation

After training, the model performance is evaluated using different classification metrics, including:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

The proposed CNN model achieved an overall classification accuracy of 98.76%, demonstrating strong capability in distinguishing benign and malignant blood cell patterns.

5.7 Prediction and Classification

The trained CNN framework is used for automated prediction and classification of microscopic blood cell images. The system accepts a blood cell image as input and predicts its corresponding class among the four target categories.

The automated classification system helps in:

- Early detection of leukemia and hematological diseases
- Reducing manual microscopic examination time
- Improving diagnostic consistency and reliability
- Supporting medical professionals in clinical decision-making

6. MAJOR FINDINGS / RESULTS

The results of the proposed research demonstrate that the Convolutional Neural Network (CNN)-based framework can effectively classify microscopic blood cell images with high accuracy and reliability. The model successfully distinguishes between benign and malignant blood cell categories, making it useful for automated hematological disease diagnosis.

Key Results

- The proposed CNN model successfully classifies blood cell images into four categories:
- Benign
- Malignant Pre-B
- Malignant Pro-B
- Malignant Early Pre-B
- The model achieved an overall classification accuracy of **98.76%**
- High values of precision, recall, and F1-score were achieved across all classes
- The confusion matrix analysis showed very low misclassification among blood cell categories

Observations

- The CNN model effectively learned important morphological features such as cell shape, texture, boundaries, and internal structures
- Malignant blood cell classes were identified with particularly high accuracy and reliability
- Training and validation accuracy curves showed stable learning performance throughout the training process
- Minimal differences between training and validation loss indicated good generalization capability and reduced overfitting

Model Performance

- The proposed CNN architecture demonstrated strong performance in multi-class blood cell classification tasks
- Precision, recall, and F1-score values remained consistently high for all blood cell categories
- The model showed reliable classification performance on unseen test data
- The automated framework provided faster and more consistent results compared to manual microscopic analysis

Classification Report Insights

- The Malignant Pre-B class achieved very high precision and recall values, showing strong classification capability
- Benign blood cell images were also classified accurately with minimal false predictions
- Macro-average and weighted-average scores above 97% confirmed balanced model performance across all classes

Visualization Insights

Different visualization techniques were used to analyze and evaluate the performance of the proposed model using Matplotlib and Seaborn. The visualizations revealed:

- Clear convergence patterns in training and validation accuracy curves
- Consistent reduction in training and validation loss across epochs
- Strong diagonal values in the confusion matrix, indicating accurate classification
- Effective differentiation between benign and malignant blood cell classes

Overall Findings

The proposed CNN-based automated blood cell classification framework proved to be highly accurate, reliable, and efficient for hematological disease diagnosis. The system reduces dependency on manual microscopic examination, minimizes human error, and supports medical professionals in early disease detection and clinical decision-making.

7. CONCLUSION AND FUTURE SCOPE

Conclusion

This research successfully demonstrates the application of deep learning techniques for automated blood cell classification using a Convolutional Neural Network (CNN)-based framework. The proposed system effectively classifies microscopic blood cell images into four clinically important categories: Benign, Malignant Pre-B, Malignant Pro-B, and Malignant Early Pre-B.

The CNN model achieved an overall classification accuracy of **98.76%**, along with high precision, recall, and F1-score values across all classes. The training and validation analyses showed stable learning performance with minimal overfitting, while the confusion matrix confirmed accurate classification with very low misclassification rates.

The proposed framework effectively learns important morphological features such as cell texture, boundaries, shape, and internal structures directly from microscopic blood cell images. Compared to traditional manual microscopic analysis, the automated CNN-based approach provides faster, more reliable, and more consistent diagnostic support while reducing human error and diagnostic workload.

The results indicate that the proposed system can serve as an efficient clinical decision-support tool for the early detection and diagnosis of hematological diseases such as leukemia. Overall, this research highlights the potential of deep learning and medical image analysis techniques in improving automated healthcare diagnostics.

Future Scope

- Use larger and more diverse clinical datasets to improve model robustness and generalization capability
- Implement advanced deep learning architectures such as ResNet, EfficientNet, and transformer-based models for enhanced classification accuracy
- Apply advanced image augmentation techniques to improve performance on imbalanced and noisy datasets
- Integrate Explainable Artificial Intelligence (XAI) methods to highlight important regions in blood cell images and improve clinical interpretability
- Develop a real-time web-based diagnostic system for automated blood cell analysis and clinical use
- Extend the framework for the detection and classification of additional hematological diseases and microscopic medical image datasets
- Integrate the proposed framework with hospital management and diagnostic systems for practical healthcare applications

8. REFERENCES

1. C. Cheuque, M. Querales, R. León, R. Salas, and R. Torres, “An Efficient Multi-Level Convolutional Neural Network Approach for White Blood Cells Classification,” *Diagnostics*, vol. 12, no. 2, p. 248, 2022.
2. K. Anita Davamani, C. R. Rene Robin, D. Doreen Robin, and L. Jani Anbarasi, “Adaptive Blood Cell Segmentation and Hybrid Learning-Based Blood Cell Classification: A Meta-Heuristic-Based Model,” *Biomedical Signal Processing and Control*, vol. 75, p. 103570, 2022.
3. M. Shahzad et al., “Blood Cell Image Segmentation and Classification: A Systematic Review,” *PeerJ Computer Science*, vol. 10, p. e1813, 2024.
4. Y. Li et al., “Domain-Invariant Representation Learning via SAM for Blood Cell Classification,” *Pattern Recognition*, vol. 170, p. 112000, 2026.
5. A. Bhattacharjee, V. Gupta, M. Kaur, S. Mehta, E. Jain, and P. Kaushik, “Blood Cell Cancer Classification Using ResNet50 and Transfer Learning: Enhancing Diagnostic Accuracy in Hematology,” 2024.
6. E. Jain and P. Kaushik, “Enhancing Precision in Blood Cell Cancer Classification: Cutting-Edge CNN Techniques for Improved Accuracy,” 2024.
7. P. Kaur et al., “Blood Cell Classification for Cancer Detection Using EfficientNetB3,” 2025.
8. P. Kaushik and P. Sharma, “Automated Leukemia Detection Using EfficientNetB3: A Robust Deep Learning Approach for Accurate and Efficient Diagnosis,” 2025.
9. P. Kaushik and P. Sharma, “Deep Learning for Automated Malaria Detection: An EfficientNetB3-Based Analysis of Microscopic Cell Images,” 2024.
10. P. Kaushik and P. Sharma, “Deep Learning-Based Malaria Detection: A Robust CNN Framework for Accurate and Automated Diagnosis,” 2025.
11. E. Jain et al., “Swin Transformer and Momentum Contrast (MoCo) in Leukemia Diagnostics: A New Paradigm in AI-Driven Blood Cell Cancer Classification,” *IEEE Access*, vol. 13, pp. 82441–82466, 2025.
12. H. Üzen and H. Firat, “A Hybrid Approach Based on Multipath Swin Transformer and ConvMixer for White Blood Cells Classification,” *Health Information Science and Systems*, vol. 12, no. 1, p. 33, 2024.
13. S. Chen, S. Lu, S. Wang, Y. Ni, and Y. Zhang, “Shifted Window Vision Transformer for Blood Cell Classification,” *Electronics*, vol. 12, no. 11, p. 2442, 2023.
14. O. M. Alshehri et al., “Dynamic Kernel Generation Through Hybrid Involution and Convolution Neural Networks for Leukemia and White Blood Cell Classification,” *Scientific Reports*, vol. 15, no. 1, p. 43844, 2025.
15. E. Jain, V. Kukreja, and P. Kaushik, “OsteoFusionFormer: Dual-Stage Transformer Fusion Framework for Knee Osteoporosis Diagnosis,” *The Knee*, vol. 61, p. 104428, 2026.
16. S. Mourya, S. Kant, P. Kumar, A. Gupta, and R. Gupta, “ALL Challenge Dataset of ISBI 2019 (C-NMC 2019),” The Cancer Imaging Archive, 2019.